

Homework 5

1. Suppose ϕ is of class C^3 on $[-1, 1]$. Show that $P.V. \int_{-1}^1 x^{-3} \phi(x) dx$ exists if and only if $\phi'(0) = 0$. (Hint: Consider the second-order Taylor expansion of ϕ .)
2. Assume that $f(x) \geq 0$ and that f decreases monotonically on $[1, \infty)$. Prove that

$$\int_1^\infty f(x) dx$$

converges if and only if

$$\sum_{n=1}^{\infty} f(n)$$

converges.

3. Let f be continuous and let g be C^1 on $[a, \infty)$. Suppose that (i) the function $F(x) = \int_a^x f(t) dt$ remains bounded as $x \rightarrow \infty$; (ii) $g'(x) \leq 0$ on $[a, \infty)$ and $\lim_{x \rightarrow \infty} g(x) = 0$. Show that $\int_a^\infty f(x)g(x) dx$ converges.
4. Suppose α increases monotonically on $[a, b]$, g is continuous, and $g(x) = G'(x)$ for $a \leq x \leq b$. Prove that

$$\int_a^b \alpha(x) g(x) dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G d\alpha.$$

5. Suppose f is a real, continuously differentiable function on $[a, b]$, $f(a) = f(b) = 0$, and

$$\int_a^b f^2(x) dx = 1.$$

Prove that

$$\int_a^b x f(x) f'(x) dx = -\frac{1}{2},$$

and that

$$\int_a^b [f'(x)]^2 dx \int_a^b x^2 f^2(x) dx > \frac{1}{4}.$$

(Hint: Holder's inequality.)